

Privacy in Statistics and Machine Learning Spring 2025

In-class Exercises for Lecture 19 (Two-player Zero-Sum Games, Part I)

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Problems with marked with an asterisk () are more challenging or open-ended.*

1. Consider a two-player zero-sum game described by a payoff matrix $M \in \mathbb{R}^{|\mathcal{R}| \times |\mathcal{C}|}$ and let $(\mathbf{r}, \mathbf{c})$ be a pair of equilibrium strategies. The support of a strategy is the set of actions with non-zero probability, so

$$\text{supp}(\mathbf{r}) = \{i : r_i > 0\}$$

and likewise for $\text{supp}(\mathbf{c})$. Prove that every i in the support of $\mathbf{r}$ is a best-response to $\mathbf{c}$. That is

$$\forall i \in \text{supp}(\mathbf{r}) \quad \mathbb{E}_{j \sim \mathbf{c}} (M_{i,j}) = \max_{i' \in \mathcal{R}} \mathbb{E}_{j \sim \mathbf{c}} (M_{i',j})$$

Note that the analogous statement (with min in place of max) will be true for all actions in the support of $\mathbf{c}$ by symmetry, but don't spend time proving it separately.

2. Consider the two-player zero-sum game with two actions for each player described by the payoff matrix

$$\begin{bmatrix} +2 & -1 \\ -2 & +3 \end{bmatrix} \tag{1}$$

Compute a pair of equilibrium strategies $(\mathbf{r}, \mathbf{c})$ for this game. [**Hint:** How does the property you proved in Question 1 help you find the equilibrium?]

3. The minmax theorem is the basis of one of the most widely used approaches for proving *lower bounds* on the performance of algorithms. Suppose we want to design an algorithm A that takes an input x from some finite set $\mathcal{X}$ and computes something about x . We have some real-valued *cost function* $\text{cost}(x, A)$ describing the cost of running A on input x . (This cost could be running time, space usage, query complexity, some measure of "error". It doesn't really make a difference.) For a given algorithm A , the *worst-case cost* of A is

$$\max_{\text{inputs } x} \text{cost}(x, A)$$

A randomized algorithm R can always be viewed as a distribution over deterministic algorithms. In that case, we can consider the *worst-case expected cost* of R as

$$\max_{\text{inputs } x} \mathbb{E}_{A \sim R} (\text{cost})(x, A)$$

- (a) (Yao's minimax principle, part 1) Show that if there exists a distribution $\mathbf{x}$ on inputs such that for every algorithm A

$$\mathbb{E}_{x \sim \mathbf{x}} (\text{cost}(x, A)) \geq T$$

then for every randomized algorithm A , the worst-case expected cost is at least T .

- (b) (Yao's minimax principle, part 2) Show that if every randomized algorithm R has worst-case expected cost at least T , then there exists a distribution on inputs $\mathbf{x}$ such that for every algorithm A

$$\mathbb{E}_{\mathbf{x} \sim \mathbf{x}} (\text{cost}(\mathbf{x}, A)) \geq T$$

In other words, if every (randomized) algorithm has to have high cost on some input, then there is a single distribution on inputs such that every (randomized) algorithm has to have high expected cost on that distribution.

Note: You can assume that the set of algorithms is finite. For example, they could be defined by Boolean circuits of some finite size.

[*Hint:* Consider a game in which the row player chooses an algorithm and the column player chooses an input.]

4. In a previous lecture we showed that MWEM is (ϵ, δ) -differentially private and can answer a set of k queries on a dataset in $\mathcal{U}^n$ with error at most $\leq \alpha$ on every query (with high probability), provided that

$$n \gtrsim \frac{(\log |\mathcal{U}|)^{1/2} (\log \frac{1}{\delta})^{1/2} (\log k)}{\epsilon \alpha^2}$$

Modify the analysis of the algorithm to ensure $(\epsilon, 0)$ -differential privacy? Prove a similar guarantee to above, showing that the algorithm is accurate provided that

$$n \gtrsim \frac{(\log |\mathcal{U}|)^a (\log k)^b}{\epsilon^c \alpha^d}$$

for some constants a, b, c, d . What parts of the algorithm and its analysis have to change?